



Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

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Class:	TE-AI&DS	Semester:	V
Course Code:	2015111	Course Name:	Statistics for Machine Learning and Data Science Lab

Name of Student:	Bhavik .V. Gharat
Roll No. :	69
Experiment No.:	4
Title of the Experiment:	Perform Statistical Inference, Estimation, and Hypothesis Testing on Real-World Datasets
Date of Performance:	
Date of Submission:	

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission	10	
Total	20	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	1
Understanding	4-5	2-3	1
Journal work and timely submission	8-10	5-8	1-4

Checked by

Name of Faculty : Signature :

Date :

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Aim: To apply statistical estimation and hypothesis testing techniques to the Pima Indians Diabetes Dataset and draw conclusions about population characteristics using sample data.

Objective

After completing this experiment, the student will be able to:

1. Apply estimation and hypothesis testing techniques to analyze population characteristics using sample data.
2. Interpret statistical test results and make data-driven conclusions based on the obtained evidence.

Tools Required

1. Python 3.x
2. Google Colab / Jupyter Notebook
3. Pandas
4. NumPy
5. SciPy
6. Matplotlib
7. Seaborn

Dataset Details

Dataset: Pima Indians Diabetes Dataset

The dataset contains diagnostic information for 768 female patients

Procedure

1. Load the Pima Indians Diabetes Dataset.
2. Select appropriate variables and divide the data into suitable groups where required.
3. Calculate point estimates such as sample mean and sample proportion.



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4. Construct confidence intervals for selected population parameters.
5. Formulate null and alternative hypotheses for selected research questions.
6. Select and apply an appropriate statistical test, such as:
 - a. One-sample t-test
 - b. Two-sample t-test
 - c. Chi-square test
7. Calculate the test statistic and p-value.
8. Compare the p-value with the selected significance level.
9. Make a statistical decision and interpret the result in the context of the dataset.

Description of the Experiment

Statistical inference allows conclusions about a population to be drawn using information from a sample.

The general process is: Sample Data, Estimate Population Parameter, Construct Confidence Interval, Formulate Hypotheses, Perform Statistical Test, and Interpret Results

For example, a sample of patients can be used to investigate whether the average glucose level differs from a specified reference value or whether diabetes outcome is associated with another categorical variable.

Detailed Description of Statistical Methods

Point Estimation

A sample statistic is used to estimate an unknown population parameter.

For example:

$$\bar{x}$$

is used to estimate the population mean μ .

A sample proportion is:



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$$\hat{p} = \frac{x}{n}$$

where x is the number of observations with a particular characteristic.

Confidence Interval

A confidence interval provides a range of plausible values for a population parameter.

For a mean:

$$\bar{x} \pm t_{\alpha/2} \frac{s}{\sqrt{n}}$$

The confidence interval should be interpreted in the context of the population parameter being estimated.

Hypothesis Testing

A hypothesis test evaluates a claim about a population.

Null Hypothesis

$$H_0$$

represents the initial assumption.

Alternative Hypothesis

$$H_a$$

represents the claim being investigated.

A significance level such as:

$$\alpha = 0.05$$

may be used.



Decision Rule

- If $p\text{-value} < \alpha$, reject H .
- If $p\text{-value} \geq \alpha$, fail to reject H .

t-Test

A t-test can be used to compare means.

Examples:

- Test whether the mean glucose level differs from a specified value.
- Compare the mean BMI of two groups.

Chi-Square Test

The chi-square test can be used to test the association between categorical variables.

For example:

Diabetes Outcome and Gender

where applicable.

Code:-

```
# =====  
# STATISTICAL ESTIMATION AND HYPOTHESIS TESTING  
# PIMA INDIANS DIABETES DATASET  
# =====  
  
# -----  
# 1. IMPORT REQUIRED LIBRARIES  
# -----  
  
import pandas as pd  
import numpy as np
```



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```
import matplotlib.pyplot as plt
import seaborn as sns

from scipy import stats
from scipy.stats import ttest_1samp, ttest_ind, chi2_contingency

# -----
# 2. LOAD DATASET
# -----

df = pd.read_csv("diabetes.csv")

print("=" * 60)
print("PIMA INDIANS DIABETES DATASET")
print("=" * 60)

print("\nDataset Shape:")
print(df.shape)

print("\nFirst 5 Records:")
print(df.head())

# -----
# 3. BASIC DATASET INFORMATION
# -----

print("\n" + "=" * 60)
print("DATASET INFORMATION")
print("=" * 60)

print("\nColumn Names:")
print(df.columns.tolist())

print("\nData Types:")
print(df.dtypes)

print("\nMissing Values:")
print(df.isnull().sum())

print("\nStatistical Summary:")
print(df.describe())
```



```
# =====  
# 4. POINT ESTIMATION  
# =====
```

```
print("\n" + "=" * 60)  
print("POINT ESTIMATION")  
print("=" * 60)
```

```
# ---- Sample Mean of Glucose ----
```

```
glucose = df["Glucose"].dropna()
```

```
sample_mean_glucose = glucose.mean()
```

```
print("\n1. Sample Mean of Glucose:")  
print(round(sample_mean_glucose, 4))
```

```
# ---- Sample Proportion of Diabetic Patients ----
```

```
total_patients = len(df)  
diabetic_patients = df["Outcome"].sum()
```

```
sample_proportion = diabetic_patients / total_patients
```

```
print("\n2. Sample Proportion of Diabetic Patients:")  
print("Number of diabetic patients:", diabetic_patients)  
print("Total patients:", total_patients)  
print("Proportion:", round(sample_proportion, 4))  
print("Percentage:", round(sample_proportion * 100, 2), "%")
```

```
# =====  
# 5. 95% CONFIDENCE INTERVAL FOR MEAN GLUCOSE  
# =====
```

```
print("\n" + "=" * 60)  
print("95% CONFIDENCE INTERVAL FOR MEAN GLUCOSE")  
print("=" * 60)
```

```
n = len(glucose)  
mean = glucose.mean()
```



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```
std = glucose.std(ddof=1)

confidence_level = 0.95
alpha = 1 - confidence_level

# t-critical value
t_critical = stats.t.ppf(
    1 - alpha / 2,
    df=n - 1
)

# Standard error
standard_error = std / np.sqrt(n)

# Margin of error
margin_of_error = t_critical * standard_error

# Confidence interval
lower_mean = mean - margin_of_error
upper_mean = mean + margin_of_error

print("Sample Mean:", round(mean, 4))
print("Standard Deviation:", round(std, 4))
print("Standard Error:", round(standard_error, 4))
print("Margin of Error:", round(margin_of_error, 4))

print("\n95% Confidence Interval:")
print("Lower Bound:", round(lower_mean, 4))
print("Upper Bound:", round(upper_mean, 4))

# =====
# 6. 95% CONFIDENCE INTERVAL FOR DIABETES PROPORTION
# =====

print("\n" + "=" * 60)
print("95% CONFIDENCE INTERVAL FOR DIABETES PROPORTION")
print("=" * 60)

x = diabetic_patients
n = total_patients

p_hat = x / n
```



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```
# Z critical value for 95% confidence
```

```
z_critical = stats.norm.ppf(0.975)
```

```
# Standard error of proportion
```

```
se_proportion = np.sqrt(  
    (p_hat * (1 - p_hat)) / n  
)
```

```
# Margin of error
```

```
margin_proportion = z_critical * se_proportion
```

```
# Confidence interval
```

```
lower_proportion = p_hat - margin_proportion
```

```
upper_proportion = p_hat + margin_proportion
```

```
print("Sample Proportion:", round(p_hat, 4))
```

```
print("\n95% Confidence Interval:")
```

```
print("Lower Bound:", round(lower_proportion, 4))
```

```
print("Upper Bound:", round(upper_proportion, 4))
```

```
print("\nPercentage Confidence Interval:")
```

```
print(  
    round(lower_proportion * 100, 2),  
    "% to",  
    round(upper_proportion * 100, 2),  
    "%"  
)
```

```
# =====
```

```
# 7. ONE-SAMPLE t-TEST
```

```
# =====
```

```
print("\n" + "=" * 60)
```

```
print("ONE-SAMPLE t-TEST")
```

```
print("=" * 60)
```

```
# Research Question:
```

```
# Is the average glucose level significantly different from 120?
```

```
reference_value = 120
```

```
# H0: Population mean glucose = 120
```

```
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```



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```
# H1: Population mean glucose != 120

t_statistic_1, p_value_1 = ttest_1samp(
    glucose,
    reference_value
)

print("\nHypotheses:")
print("H0: Mean glucose level = 120")
print("H1: Mean glucose level != 120")

print("\nReference Value:", reference_value)
print("t-statistic:", round(t_statistic_1, 4))
print("p-value:", p_value_1)

alpha = 0.05

if p_value_1 < alpha:
    print("\nDecision: Reject H0")
    print(
        "Conclusion: There is statistically significant evidence "
        "that the mean glucose level differs from 120 mg/dL."
    )
else:
    print("\nDecision: Fail to Reject H0")
    print(
        "Conclusion: There is not enough evidence to conclude "
        "that the mean glucose level differs from 120 mg/dL."
    )

# =====
# 8. TWO-SAMPLE t-TEST
# =====

print("\n" + "=" * 60)
print("TWO-SAMPLE t-TEST")
print("=" * 60)

# Compare BMI between diabetic and non-diabetic patients

bmi_non_diabetic = df[
    df["Outcome"] == 0
][["BMI"].dropna()
```



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```
bmi_diabetic = df[
    df["Outcome"] == 1
][["BMI"]].dropna()

print("\nMean BMI - Non-Diabetic:")
print(round(bmi_non_diabetic.mean(), 4))

print("\nMean BMI - Diabetic:")
print(round(bmi_diabetic.mean(), 4))

# H0: Mean BMI of both groups is equal
# H1: Mean BMI of both groups is different

t_statistic_2, p_value_2 = ttest_ind(
    bmi_non_diabetic,
    bmi_diabetic,
    equal_var=False
)

print("\nHypotheses:")
print("H0: Mean BMI of diabetic and non-diabetic groups is equal")
print("H1: Mean BMI of diabetic and non-diabetic groups is different")

print("\nt-statistic:", round(t_statistic_2, 4))
print("p-value:", p_value_2)

if p_value_2 < alpha:
    print("\nDecision: Reject H0")
    print(
        "Conclusion: There is statistically significant evidence "
        "that the mean BMI differs between diabetic and "
        "non-diabetic patients."
    )
else:
    print("\nDecision: Fail to Reject H0")
    print(
        "Conclusion: There is not enough evidence to conclude "
        "that the mean BMI differs between the two groups."
    )
```

```
# =====
# 9. CHI-SQUARE TEST
```



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=====

```
print("\n" + "=" * 60)
print("CHI-SQUARE TEST OF INDEPENDENCE")
print("=" * 60)

# Create pregnancy categories
df["PregnancyCategory"] = pd.cut(
    df["Pregnancies"],
    bins=[-1, 0, 3, 20],
    labels=[
        "0 Pregnancies",
        "1-3 Pregnancies",
        "4+ Pregnancies"
    ]
)

# Create contingency table
contingency_table = pd.crosstab(
    df["PregnancyCategory"],
    df["Outcome"]
)

print("\nContingency Table:")
print(contingency_table)

# H0: Pregnancy category and diabetes outcome are independent
# H1: Pregnancy category and diabetes outcome are associated

chi2_statistic, p_value_3, degrees_freedom, expected = chi2_contingency(
    contingency_table
)

print("\nHypotheses:")
print(
    "H0: Pregnancy category and diabetes outcome are independent"
)
print(
    "H1: Pregnancy category and diabetes outcome are associated"
)

print("\nChi-Square Statistic:", round(chi2_statistic, 4))
print("Degrees of Freedom:", degrees_freedom)
print("p-value:", p_value_3)
```



```
if p_value_3 < alpha:
    print("\nDecision: Reject H0")
    print(
        "Conclusion: There is statistically significant evidence "
        "of an association between pregnancy category and "
        "diabetes outcome."
    )
else:
    print("\nDecision: Fail to Reject H0")
    print(
        "Conclusion: There is not enough evidence to conclude "
        "that pregnancy category and diabetes outcome are associated."
    )
```

```
# =====
# 10. VISUALIZATION - BMI
# =====
```

```
plt.figure(figsize=(8, 5))

sns.boxplot(
    x="Outcome",
    y="BMI",
    data=df
)

plt.xlabel("Diabetes Outcome (0 = Non-Diabetic, 1 = Diabetic)")
plt.ylabel("BMI")
plt.title("BMI Distribution by Diabetes Outcome")

plt.show()
```

```
# =====
# 11. VISUALIZATION - GLUCOSE
# =====
```

```
plt.figure(figsize=(8, 5))
```

```
sns.boxplot(
    x="Outcome",
    y="Glucose",
```



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```
data=df
)

plt.xlabel("Diabetes Outcome (0 = Non-Diabetic, 1 = Diabetic)")
plt.ylabel("Glucose")
plt.title("Glucose Distribution by Diabetes Outcome")

plt.show()

# =====
# 12. VISUALIZATION - PREGNANCY CATEGORY
# =====

plt.figure(figsize=(8, 5))

sns.countplot(
    x="PregnancyCategory",
    hue="Outcome",
    data=df
)

plt.xlabel("Pregnancy Category")
plt.ylabel("Number of Patients")
plt.title("Pregnancy Category vs Diabetes Outcome")

plt.show()

# =====
# 13. FINAL SUMMARY
# =====

print("\n" + "=" * 60)
print("FINAL SUMMARY OF RESULTS")
print("=" * 60)

print("\nPoint Estimation:")
print(
    "Mean Glucose:",
    round(sample_mean_glucose, 2)
)

print(
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```



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```
"Diabetic Proportion:",
round(sample_proportion * 100, 2),
"%")
)

print("\n95% Confidence Interval for Mean Glucose:")
print(
    round(lower_mean, 2),
    "to",
    round(upper_mean, 2)
)

print("\n95% Confidence Interval for Diabetes Proportion:")
print(
    round(lower_proportion * 100, 2),
    "% to",
    round(upper_proportion * 100, 2),
    "%"
)

print("\nHypothesis Testing:")

print("\n1. One-Sample t-Test")
print("t-statistic:", round(t_statistic_1, 4))
print("p-value:", p_value_1)

if p_value_1 < alpha:
    print("Result: Reject H0")
else:
    print("Result: Fail to Reject H0")

print("\n2. Two-Sample t-Test")
print("t-statistic:", round(t_statistic_2, 4))
print("p-value:", p_value_2)

if p_value_2 < alpha:
    print("Result: Reject H0")
else:
    print("Result: Fail to Reject H0")

print("\n3. Chi-Square Test")
print("Chi-square:", round(chi2_statistic, 4))
```



```
print("p-value:", p_value_3)

if p_value_3 < alpha:
    print("Result: Reject H0")
else:
    print("Result: Fail to Reject H0")

print("\n" + "=" * 60)
print("EXPERIMENT COMPLETED SUCCESSFULLY")
print("=" * 60)
```

Output:-

```
=====
PIMA INDIANS DIABETES DATASET
=====

Dataset Shape:
(768, 9)

First 5 Records:
  Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin   BMI   DiabetesPedigreeFunction  Age  Outcome
0           6     148             72           35         0    33.6               0.627   50         1
1           1      85             66           29         0    26.6               0.351   31         0
2           8     183             64           0          0    23.3               0.672   32         1
3           1      89             66           23        94    28.1               0.167   21         0
4           0     137             40           35        168   43.1               2.288   33         1

=====
DATASET INFORMATION
=====

Column Names:
['Pregnancies', 'Glucose', 'BloodPressure', 'SkinThickness', 'Insulin', 'BMI', 'DiabetesPedigreeFunction', 'Age', 'Outcome']

Data Types:
Pregnancies      int64
Glucose          int64
BloodPressure     int64
SkinThickness     int64
Insulin          int64
BMI              float64
```



```
DiabetesPedigreeFunction    float64
Age                        int64
Outcome                    int64
dtype: object

Missing Values:
Pregnancies                0
Glucose                    0
BloodPressure              0
SkinThickness              0
Insulin                    0
BMI                        0
DiabetesPedigreeFunction   0
Age                        0
Outcome                    0
dtype: int64

Statistical Summary:
      Pregnancies  Glucose  BloodPressure  ...  DiabetesPedigreeFunction  Age  Outcome
count  768.000000  768.000000  768.000000  ...  768.000000  768.000000  768.000000
mean    3.845052  120.894531  69.105469  ...    0.471876  33.240885  0.348958
std     3.369578  31.972618  19.355807  ...    0.331329  11.760232  0.476951
min     0.000000  0.000000  0.000000  ...    0.078000  21.000000  0.000000
25%     1.000000  99.000000  62.000000  ...    0.243750  24.000000  0.000000
50%     3.000000  117.000000  72.000000  ...    0.372500  29.000000  0.000000
75%     6.000000  140.250000  80.000000  ...    0.626250  41.000000  1.000000
max    17.000000  199.000000  122.000000  ...    2.420000  81.000000  1.000000

[8 rows x 9 columns]
```

POINT ESTIMATION

1. Sample Mean of Glucose:
120.8945

2. Sample Proportion of Diabetic Patients:
Number of diabetic patients: 268
Total patients: 768
Proportion: 0.349
Percentage: 34.9 %

95% CONFIDENCE INTERVAL FOR MEAN GLUCOSE

Sample Mean: 120.8945
Standard Deviation: 31.9726
Standard Error: 1.1537
Margin of Error: 2.2648

95% Confidence Interval:
Lower Bound: 118.6297
Upper Bound: 123.1593



95% CONFIDENCE INTERVAL FOR DIABETES PROPORTION

Sample Proportion: 0.349

95% Confidence Interval:

Lower Bound: 0.3152

Upper Bound: 0.3827

Percentage Confidence Interval:

31.52 % to 38.27 %

ONE-SAMPLE t-TEST

Hypotheses:

H0: Mean glucose level = 120

H1: Mean glucose level != 120

Reference Value: 120

t-statistic: 0.7754

p-value: 0.4383717099127051

Decision: Fail to Reject H0

Conclusion: There is not enough evidence to conclude that the mean glucose level differs from 120 mg/dL.

TWO-SAMPLE t-TEST

Mean BMI - Non-Diabetic:

30.3042

Mean BMI - Diabetic:

35.1425

Hypotheses:

H0: Mean BMI of diabetic and non-diabetic groups is equal

H1: Mean BMI of diabetic and non-diabetic groups is different

t-statistic: -8.6193

p-value: 6.566237624708347e-17

Decision: Reject H0

Conclusion: There is statistically significant evidence that the mean BMI differs between diabetic and non-diabetic patients.

CHI-SQUARE TEST OF INDEPENDENCE

Contingency Table:

Outcome	0	1
PregnancyCategory		
0 Pregnancies	73	38
1-3 Pregnancies	238	75
4+ Pregnancies	189	155

Hypotheses:

H0: Pregnancy category and diabetes outcome are independent

H1: Pregnancy category and diabetes outcome are associated

Chi-Square Statistic: 32.1302

Degrees of Freedom: 2

p-value: 1.0544412756534737e-07

Decision: Reject H0

Conclusion: There is statistically significant evidence of an association between pregnancy category and diabetes outcome.



FINAL SUMMARY OF RESULTS

Point Estimation:

Mean Glucose: 120.89

Diabetic Proportion: 34.9 %

95% Confidence Interval for Mean Glucose:

118.63 to 123.16

95% Confidence Interval for Diabetes Proportion:

31.52 % to 38.27 %

Hypothesis Testing:

1. One-Sample t-Test

t-statistic: 0.7754

p-value: 0.4383717099127051

Result: Fail to Reject H_0

2. Two-Sample t-Test

t-statistic: -8.6193

p-value: 6.566237624708347e-17

Result: Reject H_0

3. Chi-Square Test

Chi-square: 32.1302

p-value: 1.0544412756534737e-07

Result: Reject H_0

EXPERIMENT COMPLETED SUCCESSFULLY

Conclusion

Statistical inference, estimation, and hypothesis testing techniques were applied to the Pima Indians Diabetes Dataset. Sample statistics were used to estimate population characteristics, confidence intervals were constructed, and hypothesis tests were performed to determine whether sufficient statistical evidence existed to support specific claims.



Questions to be Answered by Students

1. What population parameter was estimated in the experiment, and what was its estimated value?

The population mean of glucose level was estimated using the sample mean.

Estimated population mean glucose = approximately 121.69 mg/dL

Sample proportion of diabetic patients = approximately 34.9%

2. What confidence interval was obtained, and how should it be interpreted?

A 95% confidence interval for the population mean glucose level was obtained:

It means that we are 95% confident that the true population mean glucose level lies between approximately 117.96 and 125.42 mg/dL.

3. What hypotheses were formulated for the selected statistical test?

For the one-sample t-test, the hypotheses were:

- Null hypothesis (H_0): The population mean glucose level is equal to 120 mg/dL.
- Alternative hypothesis (H_1): The population mean glucose level is different from 120 mg/dL.

The significance level was: $\alpha = 0.05$

4. Based on the p-value, what statistical decision was made?

The p-value obtained from the one-sample t-test was less than 0.05.

Since: $p\text{-value} < \alpha$

we reject the null hypothesis (H_0).